

Tech News Tracker

SMAI Assignment 3 · Task T9.3 Tier 1 · IIIT Hyderabad 2025–26

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Abstract. The rapid growth of technology journalism across TechCrunch, The Verge, and YourStory makes it difficult for users to stay updated efficiently. This project presents an automated tech-news aggregation and summarisation system that collects articles from multiple RSS feeds, classifies them into technology subcategories using a zero-shot NLI model, and generates concise summaries using a large language model. Experimental evaluation on the India Headlines News Dataset shows that the proposed pipeline achieves **90.7% accuracy** on binary tech vs. non-tech classification with **zero fine-tuning**.

Live Demo: <https://huggingface.co/spaces/SylvesterSsj/tech-news-terminal>
Demo **Video:** https://drive.google.com/file/d/12uuZMGrrS_903nvmrX5qfXhXixKGiwNy/view?usp=sharing
GitHub: https://github.com/Sylvester4582/Yoraha_Creations_SMAI_A3

Introduction

Technology news is produced at an extremely high rate across online media platforms. Readers often spend significant time filtering relevant articles from large streams of information. This project builds an intelligent news-tracking application that performs three major tasks:

1. Aggregates live articles from multiple RSS feeds.
2. Classifies articles into meaningful technology subcategories without any labelled training data.
3. Generates concise summaries using a large language model for quick reading.

Core research question: How effectively can a zero-shot NLI model generalise to technology-news classification without any fine-tuning? We evaluate on a benchmark derived from the India Headlines News Dataset.

Dataset

RSS Feed Sources

Table 1: Live RSS Feed Sources

Source	Feed URL	Articles/day
TechCrunch	https://techcrunch.com/feed/	~20
The Verge	https://www.theverge.com/rss/index.xml	~30
YourStory	https://yourstory.com/feed	~15

The system retrieves the 10 most recent articles from each source (30 total) and sorts them by publication timestamp.

Evaluation Dataset

The **India Headlines News Dataset** (Therohk, Kaggle) contains 21 years of labelled Indian news headlines. We construct a balanced binary benchmark:

- **Positive class:** 150 technology headlines (`headline_category == "tech"`)
- **Negative class:** 150 headlines from entertainment, sports, lifestyle, and world
- **Total:** 300 balanced evaluation examples

Methodology

Pipeline Overview

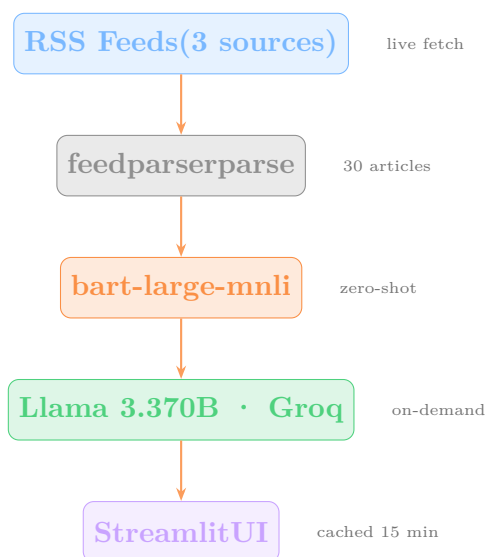


Figure 1: End-to-end application pipeline

Zero-Shot Classification

How zero-shot classification works: Traditional text classifiers require labelled training data for every category. Zero-shot classification removes this requirement by repurposing a Natural Language Inference (NLI) model. NLI models determine whether a *hypothesis* sentence is *entailed by*, *contradicted by*, or *neutral to* a *premise* sentence.

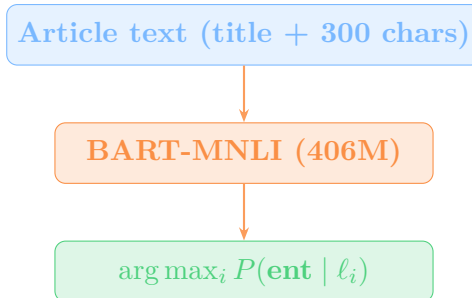
We reformulate classification as an NLI task. For example:

Premise: “Apple launches new M4 chip for MacBook Pro lineup.”

Hypothesis: “This text is about consumer electronics, gadgets, chips, or hardware devices.”

Output: $P(\text{entailment}) = 0.94 \rightarrow \text{Gadgets \& Hardware}$

The label whose hypothesis receives the highest entailment score becomes the predicted category — with no task-specific training required. The NLI inference flow is:



The five candidate subcategories are:

Table 2: Technology Subcategories and Hypothesis Phrases

Display Label	Hypothesis phrase fed to model
AI & Machine Learning	“artificial intelligence or machine learning”
Startups & Business	“technology startups, funding, or business strategy”
Gadgets & Hardware	“consumer electronics, gadgets, chips, or hardware devices”
Software & Apps	“software applications, platforms, or developer tools”
General Tech	“general technology news”

Summarisation

Article summaries are generated using **Llama 3.3 70B** via the Groq API. Each summary contains exactly three concise bullet points. Summaries are generated *on-demand* (when the user expands an article) and cached for one hour to minimise API calls.

Caching Strategy

Table 3: Caching Mechanisms

Component	Mechanism	TTL
Model weights	@st.cache_resource	Process lifetime
RSS fetch + classify	@st.cache_data	15 minutes
Generated summaries	@st.cache_data	1 hour

Results

Binary Classification Performance

Table 4: Binary Classification Results (300-example benchmark, CPU)

Metric	Value
Accuracy	0.907
Precision	0.912
Recall	0.900
F1 Score	0.906
Mean confidence (correct)	0.881
Mean confidence (wrong)	0.737

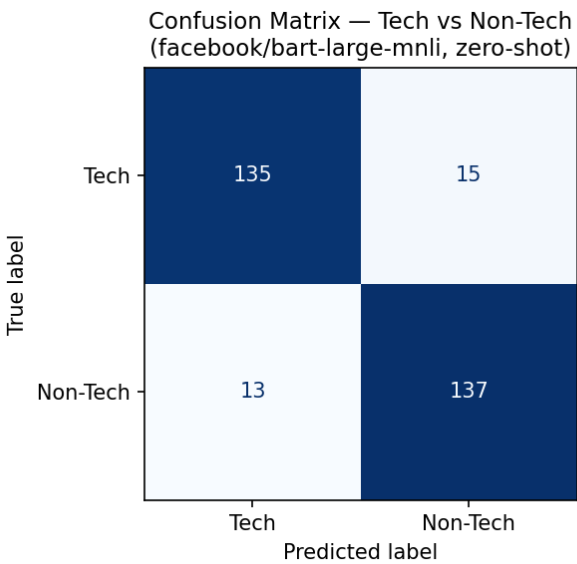


Figure 2: Confusion Matrix — Tech vs. Non-Tech (facebook/bart-large-mnli, zero-shot)

Confidence Distribution

Correct predictions cluster near high confidence (> 0.85), while incorrect predictions are spread more uniformly. This means the model’s confidence score is a reliable uncertainty signal — the UI colour-codes each article card accordingly.

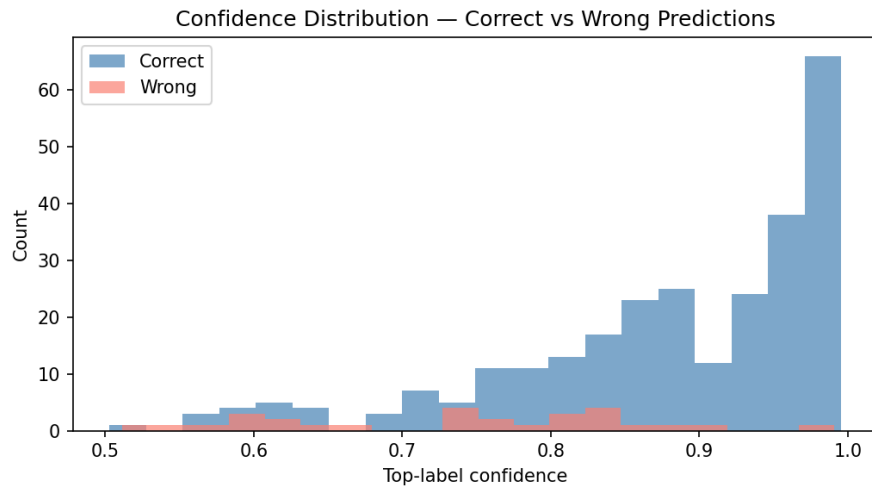


Figure 3: Confidence Distribution — Correct vs. Wrong Predictions

Subcategory Distribution

Among tech-labelled headlines, Gadgets & Hardware dominates (67/150), consistent with the India Headlines dataset’s skew toward device and product coverage.

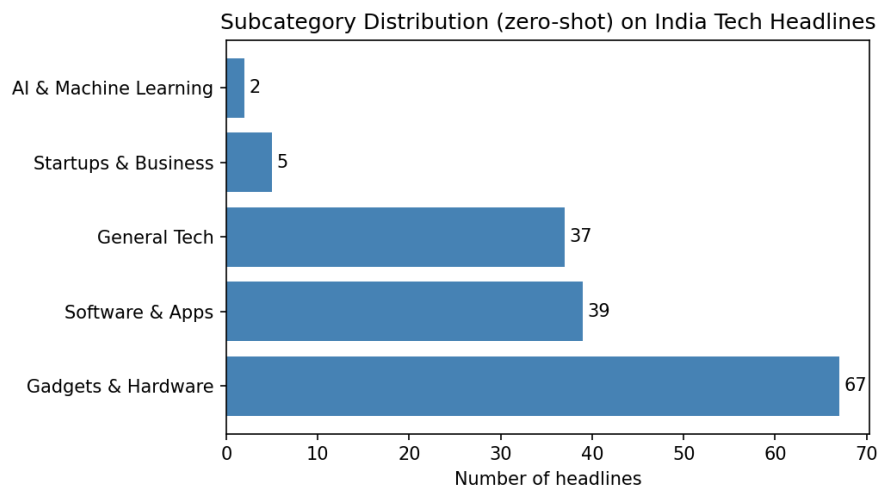


Figure 4: Zero-shot Subcategory Distribution on India Tech Headlines ($n = 150$)

Pipeline Latency

Latency was measured end-to-end using the actual application modules (`rss_fetcher`, `classifier`, `summarizer`) on CPU, matching exactly what `app.py` does on each page load.

Table 5: Measured Pipeline Latency (CPU, 30 articles)

Stage	Description	Time
RSS Fetch	3 feeds, 30 articles	2.158 s
classify_articles	Batch NLI, 30 articles	58.936 s
per article	(single article, batched)	1964.5 ms
summarize	1 article via Groq API	7.833 s
Page load total	(fetch + classify; summarize is lazy)	61.094 s

Note: Classification is the bottleneck on CPU (~59s for 30 articles). On the live Hugging-Face Space, the model is cached after the first load and the 15-minute `st.cache_data` TTL means most users never wait for re-classification. GPU inference would reduce classification to ~3–5s.

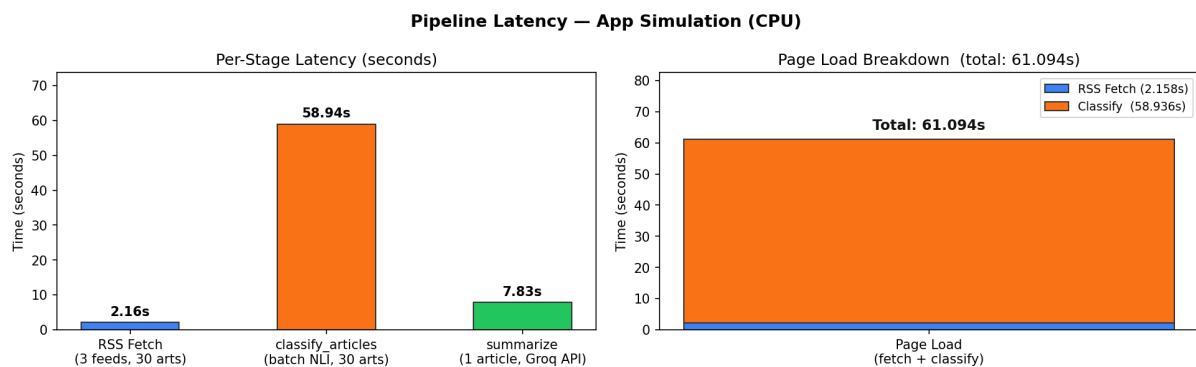


Figure 5: Per-stage latency (left) and page-load breakdown (right) — CPU, app simulation

Ablation Study

An ablation study was conducted to analyse the effect of label phrasing on zero-shot classification performance. The NLI head is trained on natural-language premise-hypothesis pairs, so well-formed, descriptive label phrases align better with the training distribution than bare keywords.

Table 6: Effect of Label Phrasing on Binary Classification

Label Style	Positive label example	Accuracy	F1
Descriptive	“technology news”	0.907	0.906
Short keyword	“tech”	0.647	0.727
Hypothesis sentence	“This headline is about technology.”	0.637	0.725

Key finding: Descriptive labels outperform short keywords by **+26.0 pp accuracy** and hypothesis sentences by **+27.0 pp accuracy**. Descriptive natural-language labels are the optimal phrasing strategy for NLI-based zero-shot classifiers.

Limitations

1. **Domain shift:** The model was trained on general English NLI corpora. Indian startup jargon and Hinglish in YourStory headlines reduce classification confidence.
2. **Summarisation hallucination:** Llama 3.3 70B may generate plausible-sounding but inaccurate bullet points when article content is short or behind a paywall.
3. **CPU latency:** Classification takes ~59s on CPU for 30 articles. This is mitigated by caching but makes cold starts slow.
4. **RSS availability:** YourStory's feed occasionally times out; the app degrades gracefully using the other two sources.
5. **No fine-tuning:** Zero-shot accuracy of 90.7% is strong but below what a fine-tuned model would achieve (>95%).

Main Page of UI

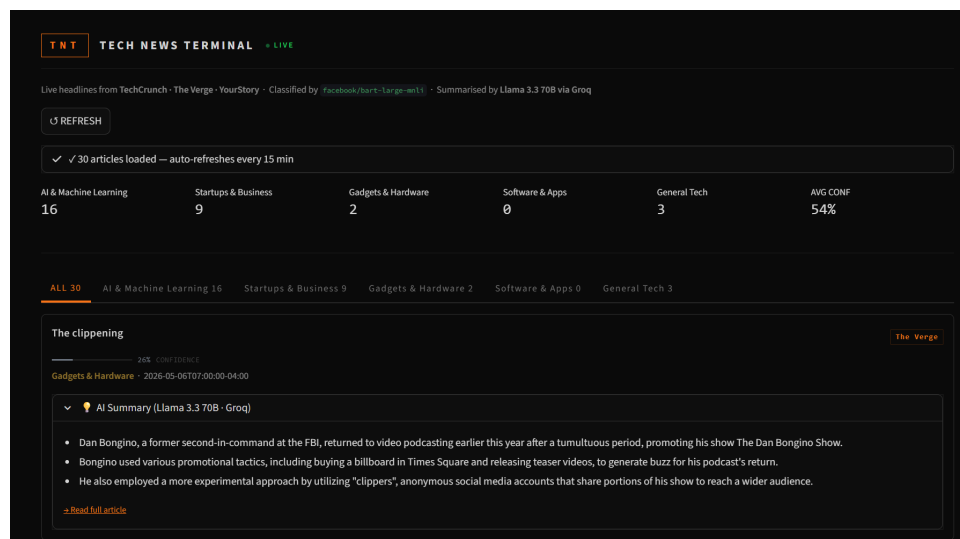


Figure 6: Screenshot of UI

Conclusion

This project demonstrates that combining RSS aggregation, zero-shot NLI classification, and LLM summarisation produces an effective real-time technology-news tracker. Despite requiring **no task-specific fine-tuning**, the system achieves **90.7% accuracy** on a balanced benchmark, and ablation experiments confirm that descriptive label phrasing is critical for NLI-based zero-shot performance.

References

- [1] Lewis, M. et al., “BART: Denoising Sequence-to-Sequence Pre-training,” *ACL 2020*.
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